Computer Organization and Architecture (CSL251)
LAB-4
2025-26 W
C & Assembly

Instructions:

1. Write separate RISC-V assembly programs for each question.
2. Use the RISC-V GNU toolchain (riscv64-unknown-elf-gcc) to assemble and link your code.
3. You can compile and generate a single executable from 2 source files using riscv64-unknown-elf-gcc lab4q1.c lab4q1.s
4. Run your program using the Spike RISC-V simulator.
5. Zip the files (lab4q1.s, lab4q1.c, lab4q2.c, lab4q2.s) and submit it with the name <ROLLNUMBER_NAME_LAB4.zip>

Question 1 (40 Marks)

1. Write an assembly code (in file lab4q1.s) that calls the functions defined in C (in file lab4q1.c).
2. Define 2 separate functions in C, One which does not take any arguments but return your jee mains rank and call that C function from assembly.
3. Declare a **global variable** named “roll” uninitialised in C and initialised in assembly.
4. Declare a second function named “print” in C that takes 3 arguments first_name, last_name (initialise them in assembly code) rank.
5. Call the “print” function (of C) from assembly by passing the necessary arguments to display the above output.

Final Output must look like:

First Name: <your_first_name>
Last Name: <your_last_name>
Roll number: <your_roll_number>
Rank: <your_jee_mains_rank>

Hints:

- Declare only ‘main’ in assembly
- Use ‘.asciz’ to initialize the strings in assembly

Question 2 (60 Marks)

1. Write a C code (in file lab4q2.c) that calls a function in assembly code (in file lab4q2.s).
2. Write a function named “concat” in the assembly that concatenates 2 strings and stores it in the data section, and returns the length of the concated string.
3. The C code should invoke this function by passing 2 arguments (string1, string2) and
4. print the concatenated string by accessing it from the assembly code data section.

● Example output:

string 1: Deadline is
string 2: today itself!!!
concatenated string: Deadline is today itself!!!, length=27

Hints:

- Use `extern` keyword

Bonus Question:

● Write the Matrix Multiplication function in assembly to call it from C.
● Initialise int array matrices (matrix_a [i ✕ j], matrix_b [j ✕ k], matrix_c [i ✕ k]) in C and pass them as arguments to the assembly function.
● Print the result matrix_c from C.

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